Group Theory Notes

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Outline

Introduction

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Python

We will use @Python@ in this course to perform mathematical computations. For the first part of the computer science course, we will look at how to use Python to work with groups.

A simple group in Python

Suppose we want to model the group from Exercise 1.1 of the previous class. Then we will need to specify the *objects* and the *actions* of the group. We can use lists to hold the objects and functions to represent the actions.

```
# group from 1.1 has as objects arrangements of a
    penny and a dime
arrangement1 = ['penny', 'dime']
arrangement2 = ['dime', 'penny']

# our action takes arrangement1 and turns it into
    arrangement2, and
# vice-versa
def group_action(arrangement):
    if arrangement[0] == 'penny':
        return ['dime', 'penny']
    else:
        return ['penny', 'dime']
```

```
None
```

Group actions in Python

Find the result of performing the group action twice in a row using the code above.

Answer: We can find the answer by computing `group_action(group_action(arrangement))` for the various arrangements.